

Research article

A Multi-Sensor IoT Platform for monitoring medicine storage beyond the hospital

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ABSTRACT

Ambient-sensitive pharmaceutical products degrade when they are exposed to unsuitable environmental conditions such as inappropriate temperature, humidity, light, or mechanical stress. While hospitals ensure proper storage, these conditions are often compromised once medications reach ambulatory patients. This paper presents a novel Internet of Things (IoT)-based platform architecture to monitor drug conservation beyond hospital facilities, extending into patients' homes. The proposed system comprises a *monitoring network* and a *servers system*. The *monitoring network* includes sensor nodes placed within the original outer packaging of the medication, capturing key environmental variables, and patient-installed gateways that locally process and transmit data to the *servers system* for near real-time storage and analysis. The *servers system* detects preservation issues and generates alerts, notifying both patients and hospital pharmacy professionals to enable timely corrective actions. Locally, the gateway warns patients of suboptimal storage conditions, allowing immediate intervention. This platform fosters collaboration between patients and pharmacists, ensuring effective follow-up on drug conservation and administration. Special consideration has been given to energy efficiency to extend sensor node battery life, making the system practical for real-world deployments. The implemented sensor node collects data every 10 s to be transmitted to the gateway, resulting in an estimated battery life of 17.5 months—sufficient for long-term operation, without frequent maintenance. By enabling continuous drug quality monitoring beyond hospitals, including at patients' homes, this platform supports safer and more effective drug therapies. This work represents a significant advancement in leveraging IoT technologies to improve medicine conservation in decentralized healthcare settings.

1. Introduction

The quality of medicines is a prerequisite for safe and effective medical treatment. Ensuring proper storage and handling of pharmaceutical products throughout the entire chain of storage, transport, and use is essential to maintaining their efficacy. In this

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regard, the incorporation of information and communication technologies (ICT) in healthcare presents an opportunity to enhance care processes while promoting active patient involvement in remote monitoring. It is important to note that patients, as end users, are often directly responsible for storing medications in their homes using available household resources.

Furthermore, according to current regulations [1], precise control of the entire cold chain is required for the storage and transport of drugs, including when medications are kept at home until administration. After discharge from the hospital, patients are typically prescribed pharmacological treatments, with hospital pharmacy services dispensing medications to ensure proper dosage adherence. These medications are stored in patients' homes, often using non-medical refrigeration devices. While hospital pharmacists provide guidance on the appropriate storage conditions and potential consequences of improper preservation, several studies [2–4] suggest that out-of-hospital medication quality management can be compromised by various factors.

Environmental conditions – particularly temperature excursions, humidity, and exposure to light – can severely affect the stability and efficacy of thermolabile or photosensitive drugs. For example, insulin loses potency when stored outside the temperature range [5]; certain vaccines may become ineffective after brief exposure to ambient temperatures; and some antibiotic suspensions degrade rapidly in humid environments. These risks are especially pronounced in domestic settings, where refrigeration devices are not designed for medical use and may exhibit temperature fluctuations, power failures, or frequent door openings. Moreover, unlike hospitals, patient homes lack continuous monitoring systems that can detect and report deviations in real time.

Existing research [6,7] highlights the need for new methods to ensure the integrity of medicines within the hospital cold chain, typically managed by healthcare professionals. However, implementing these protocols becomes challenging when patients are responsible for optimal storage, as medications are stored outside controlled hospital environments. In this context, the development of non-intrusive and user-friendly monitoring systems is crucial to safeguard therapeutic efficacy and reduce health risks associated with degraded pharmaceuticals.

Currently, medication quality control during transport between pharmaceutical companies and hospital pharmacy services, as well as within hospital facilities, relies on automated temperature monitoring systems. The same approach has been used to assess thermal profiles in the preservation of thermolabile drugs at patients' homes. Specifically, data loggers record temperature fluctuations during transport and storage, but these commercial meters are typically passive single-use devices, leading to high costs per medication. Additionally, stored data can only be accessed once the device is retrieved and analyzed, preventing real-time corrective actions. Prospective studies [5,8,9] indicate that medications often do not remain within recommended temperature ranges in home settings. Moreover, research [10] has shown that domestic refrigerators frequently exceed 8 °C or drop below 2 °C, making them unsuitable for pharmaceutical preservation.

While several IoT-based solutions have been proposed for monitoring medication storage conditions, these are often limited to centralized environments such as hospital storage rooms or transport containers. A recent review of the literature revealed a lack of platforms that provide decentralized, end-to-end monitoring extending into patients' homes. This gap highlights the need for a more comprehensive and patient-centered approach.

In response to these challenges, this work proposes a platform architecture for monitoring medicine conservation beyond hospital facilities, extending into patients' homes by leveraging Internet of Things (IoT) technologies [11]. The objective is to ensure proper conservation of ambient-sensitive medications through a collaborative approach involving both patients and pharmacy professionals to prevent the administration of degraded drugs. Utilizing remote monitoring technologies, the platform keeps patients informed while enabling effective follow-up by healthcare professionals.

The proposed system measures environmental variables affecting drug quality using sensor nodes physically attached to medication packages. These sensors periodically transmit data to a gateway device installed in the patient's home. Through this gateway, drugs can be remotely monitored while local processing is performed before securely transmitting the data to central servers. Additionally, the system autonomously alerts patients to improper storage conditions, allowing them to take corrective actions. Hospital pharmacy professionals also review the collected data and local warning behavior to ensure proper conservation and administration practices.

The scope and main contributions of the present work can be summarized as follows. Building upon previous developments of the proposed *Multi-Sensor IoT Platform*, this study introduces several key advancements: (1) integration of both commercial and custom-designed sensor nodes, enabling multi-variable environmental monitoring (temperature, humidity, UV radiation, and mechanical stress); (2) improvements to the previously deployed hardware and software architecture, enhancing its robustness and adaptability for both near real-time and retrospective data handling across heterogeneous communication protocols; (3) validation through a real-world pilot deployment involving 50 patients; and (4) insights into system usability, energy efficiency, and deployment scalability. These contributions represent a substantial step forward in making personalized, distributed medicine conservation viable in practical healthcare scenarios.

The manuscript first discusses existing approaches and limitations in the field of pharmaceutical monitoring. We then introduce the QChainMED platform, detail its architecture and implementation, and present experimental validation results. The paper concludes with a discussion and final remarks.

2. Literature review

In this section, we provide an overview of recent developments in IoT technologies and their applications across various supply chain sectors, focusing on the pharmaceutical industry and remote drug monitoring. The section is organized into three main subsections: the first discusses the integration of IoT in supply chain management, outlining its role in operational efficiency and sustainability. The second subsection reviews specific applications of IoT in pharmaceutical supply chains, including management

of cold chain conditions and remote monitoring of drugs. Finally, the third subsection presents the QChainMED platform, a solution aimed at monitoring medicine storage conditions in patients' homes. Through this organization, we provide a structured overview of the evolving role of IoT in supply chain and healthcare contexts.

IoT-Enabled Supply Chain Management

Supply Chain Management (SCM) involves monitoring and optimizing all aspects of the supply chain stages for goods and services. The Internet of Things (IoT) has contributed to modern SCM by automating and digitalizing processes, which can help reduce operating costs, improve efficiency, and enhance safety and security measures. IoT devices play an important role in supply chains by enabling real-time tracking and tracing of shipments. Taj et al. present a systematic literature review in [12] on IoT-based SCM, analyzing application domains, technologies, sensors, and devices used in such systems. This review indicates that not only IoT but also IoT-enabling technologies like artificial intelligence (AI), blockchain, and cloud computing have been recently applied in this domain and may be further explored to enhance the robustness and reliability of IoT-based SCM. Additionally, the review shows that most research focuses on the agri-food supply chain [13] and the medical and healthcare supply chain.

The role of IoT in supply chain performance is examined in [14] from a Circular Economy (CE) perspective across the entire supply chain. The findings highlight the role of IoT in real-time tracking, inventory management, and resource optimization, and identify the integration of internet-enabled processes as a contributing factor to enhancing supply chain efficiency and responsiveness.

A comprehensive exploration of Sustainable Supply Chain Management (SSCM) within the medical industry is provided in [15], discussing its role in balancing environmental responsibility, social ethics, and economic viability. Liu et al. in [16] propose a conceptual framework for SSCM under CE principles incorporating Industry 4.0 technologies. Similar to [12], this proposal integrates cloud services, AI, blockchain, and IoT, and further includes big data analytics into SSCM. This study suggests that IoT plays a role in connecting the digital and physical worlds within the supply chain. Additionally, IoT is utilized to link smart medicine shelves in factories, drug stores, and hospitals, supporting timely replenishment of inventory.

The research outlined in [17] investigates the implementation of cyber-physical smart contracts for compliance monitoring in supply chain markets. It presents a review of existing literature, outlining underlying architectures and their assessments. Cyber-physical systems (CPS) can be understood as integrations of cyber capabilities (e.g., computation and communication) with aggregations of hardware, often comprising IoT devices, to enable intelligent monitoring and control of the physical world. CPSs are defined as the combination of physical components, computing systems, and networks within larger systems, supporting intelligent monitoring and control. Smart contracts function as software systems that help manage, automate, and regulate process execution, responding to breaches and enforcing predefined terms and conditions. Many such applications operate within CPSs, adopting architectures based on IoT technologies to support monitoring and data integrity.

IoT in Pharmaceutical Supply Chains and Remote Drug Monitoring

A review on the application of IoT in pharmaceutical manufacturing, warehousing, and supply chain management is provided in [18]. It discusses how the technology can contribute to improving production efficiency, ensuring product quality, and maintaining regulatory compliance, while acknowledging challenges related to data management, privacy, and security. Each manufacturing stage, from granulation to packaging, undergoes continuous monitoring to track environmental conditions such as humidity, light, and temperature according to regulations. This real-time monitoring supports timely detection and mitigation of issues, which can help preserve product quality. IoT technologies are used to enhance the effectiveness and accuracy of warehouse processes such as stock-taking, product sorting, and inbound/outbound logistics. Smart warehouses utilize real-time information, allowing managers to monitor storage conditions, track sensitive medicines, and address issues as they arise. IoT contributes to supply chain processes by enabling new revenue opportunities and operational efficiencies, including asset tracing, operator relations, inventory prediction, connected fleets, and maintenance scheduling. IoT-enabled packaging and GPS-enabled vehicles help maintain product quality during transit. IoT systems provide real-time updates on inventory, supporting planning and coordination, potentially reducing costs and improving operations. Smart pharmaceutical packaging, supported by IoT technologies, enables efficient drug tracking throughout the pharmaceutical supply chain. These systems provide a digital record that enhances visibility and efficiency, supports product authenticity, and allows continuous monitoring of environmental conditions in cold chains. Managing cold chain conditions is important for pharmaceuticals, particularly biologics and high-value drugs that require specific temperature ranges for stability. Environmental sensors embedded within product packages monitor and update temperature data regularly, generating alerts if conditions fall outside specified ranges. This monitoring helps reduce the risk of drug spoilage due to temperature fluctuations, potentially decreasing wastage and supporting regulatory compliance. The adoption of advanced IoT technologies has the potential to impact the pharmaceutical industry by enabling innovative solutions and improving operational efficiency, contributing to the delivery of safe, sustainable, and cost-effective pharmaceutical products.

Bengiovanni et al. in [19] introduce SensIC, a cost-effective IoT-based solution to ensure the safe transportation and storage of temperature-sensitive medicines, vaccines, and biological samples. The SensIC system comprises SensBox, a smart device, and a cloud infrastructure for device management. SensBox supports Bluetooth Low Energy (BLE), Sub-GHz, and WiFi communication protocols for data forwarding, and can be equipped with various sensors to enable continuous real-time monitoring of environmental parameters such as temperature and humidity. The system supports risk management by providing alerts in the event of refrigeration system malfunctions, helping to ensure traceability. The solution complies with European regulations and guidelines for the storage and handling of sensitive medical products.

Fauzi et al. [20] present an IoT-based system designed to monitor environmental conditions in a fixed drug storage room, focusing on temperature and humidity. The system employs a DHT-11 sensor connected to a NodeMCU ESP8266 microcontroller, transmitting data to platforms such as Blynk IoT and MIT App Inventor for real-time remote visualization. This approach illustrates the use of

Table 1

Comparison of pharmaceutical IoT platforms: functional coverage and monitored parameters.

Reference	Alert generation mechanism	Transport monitoring	Patient home monitoring	In-facility storage monitoring	Parameters monitored
Bengiovanni et al. [19]	Alerts for malfunctions or deviation from configured temperature ranges.	Yes	No	Yes	Temperature, humidity, motion, CO, smoke
Bhatti et al. [21]	Alerts for temperature excursions or malfunctions.	Yes	No	Yes	Temperature
Fauzi et al. [20]	Alerts when room temp/humidity deviate or storage door is opened.	No	No	Yes	Temperature, humidity, door opening
QChainMED (This work)	Alerts for sensor malfunction or deviation in temperature, humidity, UV, or mechanical stress.	Yes	Yes	Not used, but available	Temperature, humidity, UV radiation, mechanical stress

low-cost IoT technologies in pharmaceutical settings; however, its application is limited to ambient room condition monitoring at a single, centralized location. In contrast to patient-centered systems that track conservation conditions at the medication level, this solution does not monitor individual drug packages, which may reduce its suitability for distributed, personalized medication management.

The system described in [21] uses IoT devices, including electronic sensors and GPS, to monitor the distribution and temperature of medicines during transportation. It is designed to reduce medicine waste and maintain safety against environmental factors. The system incorporates temperature sensors connected to Arduino microcontrollers for real-time temperature monitoring. Data transmission is handled via the Message Queuing Telemetry Transport (MQTT) protocol, which supports scalable and reliable communication over cellular networks. Supervisors can access medicine condition data remotely through a cloud-based platform. The system provides alerts in response to temperature deviations, facilitating timely interventions to avoid medicine spoilage or loss of efficacy. Additionally, the authors propose integrating blockchain technology to secure sensor data, aiming to prevent unauthorized access or tampering and improve overall system reliability.

The study presented in [22] examines the current state of blockchain technology research in healthcare supply chains. A review of the literature indicates that blockchain has the potential to address challenges such as counterfeiting, provenance, tracking, and authenticity, thereby improving stakeholder confidence and trust. However, adoption is limited by factors including security and privacy concerns, high implementation costs, scalability issues, interoperability challenges, trust deficits, and system complexity. Addressing these barriers is important for facilitating broader adoption of blockchain solutions in healthcare supply chains.

Recent conceptual developments in the Internet of Medical Things (IoMT) highlight the significance of human-centered design and active patient involvement, as outlined in the Healthcare 5.0 paradigm [23]. This framework promotes collaborative, patient-driven health technologies that enable personalized monitoring and real-time feedback. Our platform supports this approach by engaging patients directly through local gateway alerts and environmental data feedback, encouraging a more active role in treatment adherence and medication management compared to traditional models where patients are passive recipients of care.

However, research on the use of IoT for monitoring drug dispensation from hospitals to patients for home use remains limited. To the best of our knowledge, existing IoT-based solutions for drug distribution monitoring do not offer a comprehensive approach to managing ambient-sensitive drug handling throughout the entire process, from hospital dispensing to patient administration at home.

Table 1 presents a comparative overview of existing IoT-based pharmaceutical monitoring systems. It summarizes the functional scope of each solution by detailing the types of alarms generated, the monitored environments – including transport, in-facility storage (such as hospitals, clinics, or pharmacies), and patients' homes – and the environmental parameters each system is capable of sensing.

QChainMED: A Holistic Approach to Remote Medication Monitoring

The QChainMED platform, introduced in [24], is a technological solution designed to involve patients and pharmaceutical staff in maintaining medicine quality by monitoring storage conditions at patients' homes. The platform architecture expands hospital monitoring capabilities to patients' residences, where medicines are stored until consumption. This concept was further developed in [25], which proposed attaching a commercial sensor node to each prescribed batch of medicine doses in order to monitor conservation conditions, primarily focusing on temperature. The conservation temperature range, well-defined by the pharmaceutical industry, is a critical factor in preserving drug quality. While the commercial sensor node meets many of QChainMED's requirements, monitoring additional variables such as humidity, UV radiation exposure, and mechanical stress is also important, especially for medicines containing biological components. To address this, a new QChain Node was developed and described in [26], with an evaluation of its energy consumption confirming suitability for extended monitoring within the battery's capacity.

To contextualize the capabilities of the proposed QChainMED platform, Table 1 compares its key features with those of representative state-of-the-art systems cited in the literature. While many solutions focus on in-facility storage or transport, QChainMED is distinctive in its support for personalized, patient-home monitoring, near real-time alerting, and multi-variable sensing tailored to the conservation of drugs.

This paper presents the most advanced iteration of the QChainMED platform, introducing several key innovations that build upon our earlier developments [24–26]. The main contributions of this study include:

1. **Enhanced sensing capabilities:** Integration of both commercial sensor nodes and custom-designed QChain Nodes, enabling comprehensive environmental monitoring beyond temperature to include humidity, UV radiation, and mechanical stress—factors particularly relevant to the stability of biologic drugs.
2. **Improved communication infrastructure:** Refinement of the gateway-server architecture to support enhanced scalability and heterogeneous communication protocols, specifically LoRa and WiFi, facilitating more flexible and robust data transmission in diverse deployment scenarios.
3. **Real-world system validation:** A full-scale pilot deployment in a clinical environment involving 50 patients, offering insights into system performance under actual operating conditions, including data reliability, responsiveness, and patient usability feedback.
4. **Extended energy efficiency assessment:** Comprehensive analysis of energy consumption under prolonged use, demonstrating the system's ability to sustain long-term autonomous operation within the constraints of battery-powered devices.

This implementation is currently being tested in a pilot study at the *Hospital Universitari Son Espases*¹ in Palma, Spain. The system demonstrates near real-time remote monitoring of drug conservation conditions with patient-centered alerts and healthcare oversight. By extending the monitoring capabilities from the hospital to patients' homes, this work represents a significant step toward personalized, proactive healthcare that ensures the integrity of medicines from hospital to consumption.

3. Methodology for digital platform design

The proposed digital platform design aims to effectively monitor the drug conservation quality at patients' home when it is dispensed by the hospital pharmacy staff to the patients for storing until intake. The platform is an IoT-based approach using remote automated and real-time meters that allow healthcare staff from the hospital to supervise medicine preservation parameters. Monitoring information is forwarded in near real-time from patients' homes to the platform servers using an specifically designed and versatile monitoring network. The proposed design tries to minimize the required external interventions at patients' home, while granting an autonomy of at least four months for sensor nodes.

The proposed solution consists of two main subsystems: the *monitoring network* and the *servers system*, as shown in Fig. 1. The *monitoring network* is formed by the sensor nodes, the gateway and the wireless networks infrastructures to collect temperature, humidity, light radiation and mechanical stress values to be sent to the *servers system*. The sensor nodes are expected to be attached to the medicine, inside the patient's refrigerator, while the gateway will be located somewhere in the patient's home, near enough to establish communication with the sensor. A BLE-based communication link will be used between the sensor and the gateway. Communication between the gateway and the *servers system* can be established via LoRa, if this technology is available, or via WiFi, if the patient's home has this service. Finally, the LoRa server will connect to the *servers system* through the Internet. The patient's gateway not only forwards the received sensing data from BLE to the *servers system* using LoRaWAN or WiFi protocols but also alerts the patient when an alarm is detected. Section 4.1 is devoted to the *monitoring network* detailed description.

The *servers system* is composed of two main servers. The first one, the *Data Management server*, will be responsible for collecting data from the sensor nodes as well as for communicating configuration parameters to each patient's gateway. This server is not publicly accessible from the Internet. The second server is the *App server*, which is in charge of running the user interface, accessible by sanitary agents from the hospital. This application offers a user-friendly interface for managing patient gateways, administering sensor nodes, analyzing collected data, and handling alarms by pharmaceutical professionals. This App consists of a web-based design with encrypted communication to guarantee high security levels. A detailed description of the *servers system* will be provided in Section 4.2.

3.1. Drug quality parameters

Environmentally sensitive medications are particularly vulnerable to improper storage, as several critical factors influence their quality and effectiveness. The key variables to consider include:

- **Temperature:** This is a critical variable for many medications. Some drugs are sensitive to heat and require controlled temperatures to prevent spoilage, while others may need refrigeration to maintain stability. Monitoring and controlling storage temperatures is essential to ensure medications remain within their recommended ranges.
- **Humidity:** Excessive humidity can compromise the stability of certain medications, causing degradation or alterations in their structure. Proper storage in a dry environment is crucial to avoid these effects.
- **Light Radiation:** Exposure to light, especially ultraviolet (UV) radiation, can degrade sensitive medications, leading to the breakdown of active ingredients or changes in formulation. Medications should be stored in opaque or light-resistant containers to shield them from direct sunlight or intense light sources.
- **Mechanical Stress:** Some medications are sensitive to mechanical stress, such as excessive shaking. Vigorous or prolonged shaking can cause protein denaturation or aggregate formation. Care should be taken to avoid excessive shaking, drops, or uncontrolled movement, and when necessary, controlled shaking methods should be used.

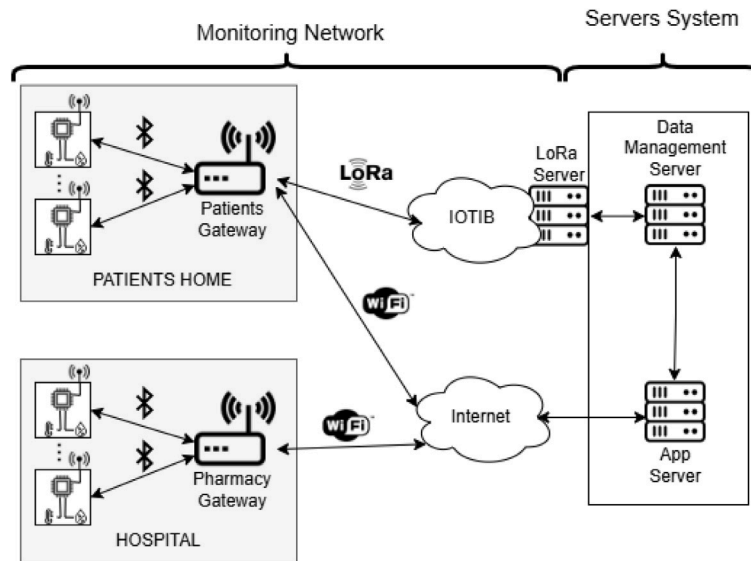


Fig. 1. Platform diagram.

Table 2
Environmental thresholds used for drug quality monitoring.

Parameter	Description	Critical limits	Adequate
Temperature (T)	Exposition to Temperature	$T > 25^{\circ}\text{C}$ or $T < 0^{\circ}\text{C}$	$2^{\circ}\text{C} \leq T \leq 8^{\circ}\text{C}$
Humidity (H)	Relative humidity level	$H > 60\%\text{HR}$	$H < 60\%\text{HR}$
Light Radiation (LR)	Exposition to light radiation	Variable	Variable
Mechanical Stress (MS)	Drop or Shaking of the drug	Variable	Variable

Table 2 summarizes the four parameters selected for this work to characterize the quality profile of medications during storage at the patient's home. It is important to note that not all parameters affect all drugs equally. In some cases, generic storage ranges can be applied, while in others, variable ranges must be tailored to the specific drug or biological substance, as detailed in **Table 2**.

The level of tolerance to light exposure varies depending on the drug and the manufacturer's specifications. The sensor used in this work measures both the maximum radiation level and the duration of light exposure, enabling precise monitoring.

Similarly, tolerance to drops or rapid shaking depends on the medication type, formulation, and specific product characteristics. No universally accepted regulations exist across regions or countries for this parameter. The proposed circuit includes functionality to measure both the maximum mechanical force and the duration of mechanical stress.

In summary, the proposed sensor measures the medication's quality profile using the four parameters outlined in **Table 2**. While well-defined exposure ranges are not established for all variables, these parameters have been identified as critical factors influencing medication quality. For the temperature parameter, in addition to the defined critical limits ($T < 0^{\circ}\text{C}$ or $T > 25^{\circ}\text{C}$), the buffer zones are also considered. Specifically, values between $0-2^{\circ}\text{C}$ or $8-25^{\circ}\text{C}$ are tolerated for short durations; alerts in these cases are only triggered if the deviation persists beyond a configurable time threshold. This mechanism avoids false alarms caused by transient fluctuations, such as brief refrigerator door openings. The collected data is monitored and reported to the proposed platform. Decisions regarding medication conservation quality are based on the analysis and interpretation of this data, which will involve input from the hospital's pharmaceutical team.

4. System architecture

4.1. Monitoring network elements

This section provides a detailed explanation of the components that constitute the *Monitoring Network*, as introduced in Section 3, with a focus on the primary functionalities of each element.

¹ <https://www.hospitalsonespases.es>.

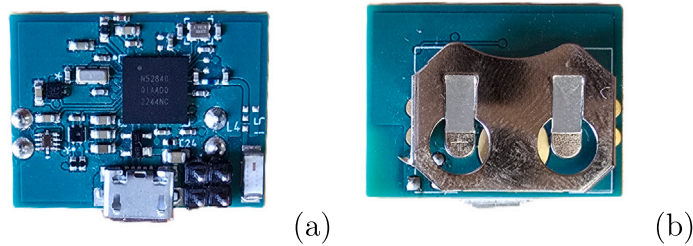


Fig. 2. Printed circuit board of sensor node. (a) The top side contains the microcontroller and the sensors. (b) The bottom side with the battery holder.

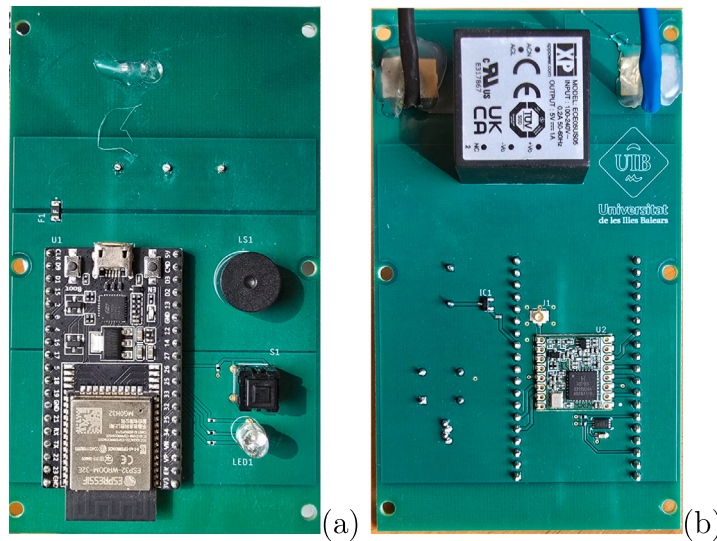


Fig. 3. Gateway board. (a) Front view with ESP32-DevKitC-32E board, buzzer, push button and RGB LED. (b) Rear view with RFM95 W LoRa module and AC-DC power trafo.

4.1.1. Sensor

A custom sensor has been developed for integration into a real IoT platform deployed in the Balearic Islands. The final PCB measures 27×14 mm with a height of 5 mm, including the CR2032 battery (220 mAh, 3 V), as shown in Fig. 2. Fig. 2a highlights key components such as the microcontroller, BLE antenna, and sensor modules. Their arrangement enables installation in direct contact with each drug unit—referring to the internal atmosphere of the primary packaging (e.g., box, blister, or bottle), not the pharmaceutical substance itself. This positioning is essential for accurate detection of parameters like light and vibration. The design supports both solid and liquid formulations, provided the packaging allows internal environmental monitoring. Sensor nodes are installed by hospital pharmacy staff and configured in *datalogger* mode to record data continuously until they connect to a patient gateway (see Fig. 1).

The sensor node integrates the nRF52840 SoC from Nordic Semiconductor, featuring a 32-bit ARM Cortex-M4 CPU with FPU, a 2.4 GHz multiprotocol transceiver, 1 MB flash, and 256 kB RAM. Sensor communication uses the I2C bus, which is disabled in low-power mode. Temperature and humidity are measured with the SHT40-AD1B-R3 (Sensirion), offering high accuracy and low power consumption (0.4 μ A active, 80 nA idle). It provides temperature readings from -40 °C to 125 °C with 0.01 °C resolution (± 0.2 °C accuracy), and humidity readings from 0%–100% RH with 0.01% resolution ($\pm 1.8\%$ RH accuracy). UV radiation is detected using the SI1147-A10-GMR (Silicon Labs), a low-power sensor with I2C interface that measures infrared proximity, UV index, and ambient light. The MC3635 accelerometer (Memsic) was selected for its ultra-low power consumption (0.3 μ A) and low noise.

4.1.2. Gateway

The gateway is a central component that connects to sensors via BLE and transmits data to a server using WiFi or LoRa. Accordingly, its hardware must support both short-range and long-range wireless communication. This design is based on the ESP32-DevKitC-32E board (Fig. 3), chosen for its integrated WiFi and BLE, open-source environment, and strong community support. LoRa capability is added via an external RFM95 W module. Additional features include a buzzer for acoustic alerts, an RGB LED for visual signals, and a push button for user interaction. Power is supplied by an ECE05US05 XP Power transformer, converting 100–240 V AC to 5 V DC, which the ESP32 uses to provide a regulated 3.3 V ($\pm 10\%$) to its components. For secure communication, the gateway incorporates an ATECC508 A cryptographic chip, as detailed in Section 5.4.

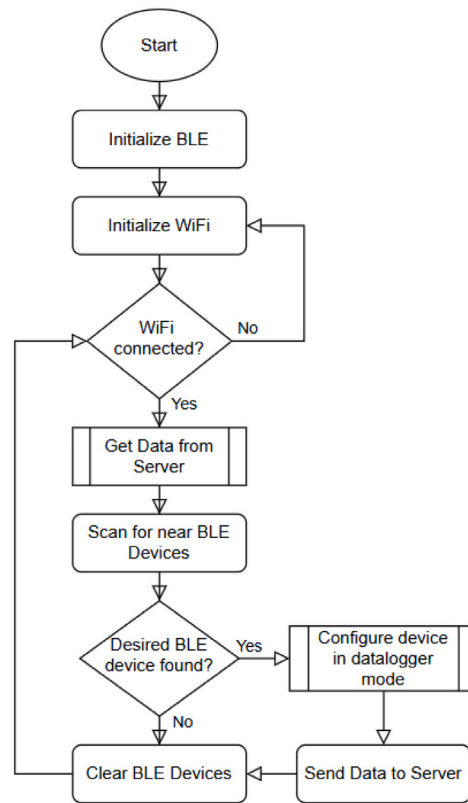


Fig. 4. Flow chart *pharmacy* gateway.

Two different types of gateways have been considered: the *pharmacy* gateway and the *patient* gateway. Although both share the same hardware, they implement different software functions and are described separately as follows.

Pharmacy gateway: *Pharmacy* gateways are used to preconfigure sensors just before delivery to patients by the hospital pharmacy (see Fig. 4). Installed in consultation rooms and powered via outlet, they run a continuous scanning loop. When patients collect multi-dose prescriptions for self-administration, they are invited to join the monitoring project. Upon consent, they are registered in the central server application and assigned both a sensor and a *patient* gateway. To monitor conditions during transport, the *pharmacy* gateway configures the sensor in *datalogger* mode, enabling continuous recording of environmental data (e.g., temperature, humidity) every 10 s into non-volatile memory. Upon arrival at the patient's home, the *patient* gateway detects the sensor, downloads the transport data, and switches it to real-time transmission mode. From then on, the sensor continues sampling every 10 s, while the *patient* gateway aggregates measurements into 10 min summaries (minimum, maximum, and average) for server transmission.

Patient gateway: The device responsible for receiving, processing, and transmitting sensor data to the central server is the *patient* gateway (see Fig. 5). It is intended to be installed in the patient's home, preferably in a power outlet near the refrigerator where medications and the attached sensor are stored.

The *patient* gateway periodically receives data from the sensor via BLE and processes it to generate local alerts for the patient in case of risk. As noted, it also transmits processed data to the central server via WiFi or LoRa, making it the link between sensor and server. One may wonder why LoRa or WiFi is not integrated directly into the sensor. The answer lies in energy efficiency: long-range transmissions significantly increase power consumption, especially during connection events. Since the sensor is battery-powered and must operate throughout the drug's monitoring period, short-range BLE communication is preferred. In contrast, the gateway, being mains-powered, has no such limitations. Therefore, its role as an intermediary – bridging low-power BLE with long-range transmission – is essential.

4.1.3. LoRa server/WiFi

This server is provided by IoTIB [27], a public LoRa network managed by the Balearic government. The LoRa server facilitates communication between the gateway and the servers system. To achieve this, the data exchange consists of two segments: one between the gateway and the LoRa server itself, and another between the LoRa server and the servers system. Two types of messages

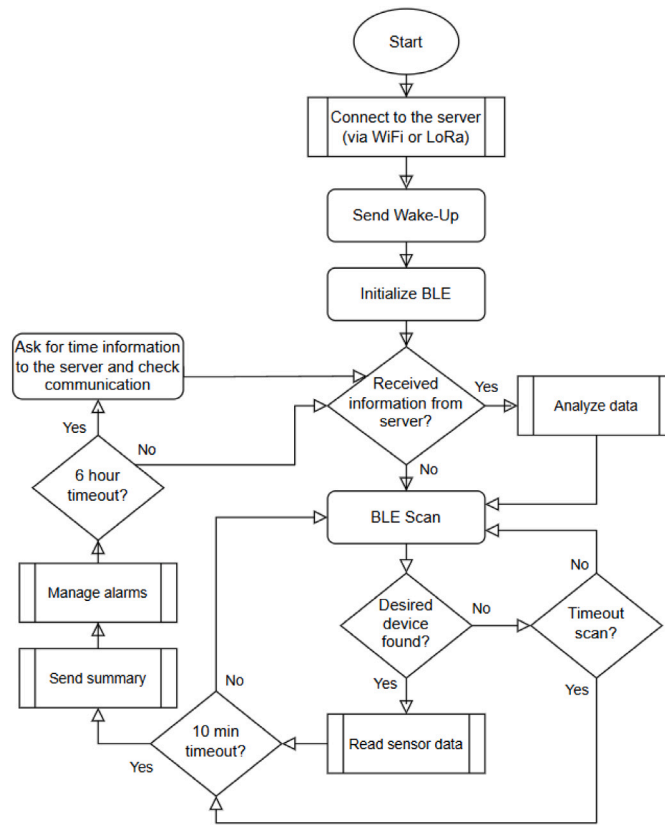


Fig. 5. Flow chart *patient gateway*.

can be distinguished in LoRa networks: *uplink frames* (data from the gateway to the server) and *downlink frames* (data from the server to the gateway). Their exchange is performed via its API (Application Programming Interface). Although it is a public service, credentials are required to use the API.

4.2. Servers system architecture

In this section, the *servers system*, previously introduced in Section 3, is described in detail, outlining the main functionalities of each server within the proposed platform.

As depicted in Fig. 1, the *servers system* comprises two components. The primary server is the *Data Management server*, which hosts a database and processes all collected data. The *App server* hosts the application interface for accessing this data. Both servers operate on the Ubuntu Server operating system.

4.2.1. App server

This server hosts the platform's web App, developed with Next.js [28]. This App serves various purposes, ranging from managing users to monitoring the medicine status. The App distinguishes between two user profiles:

- **Pharmacy specialist:** Users in this profile can add new patients to the App, deliver medicines with a sensor attached to them, monitor the conservation status of the medicines once delivered to the patient, and send alerts to the gateways if needed.
- **Admin:** In addition to all the functions of the *Pharmacy specialist*, these users can also create, delete, or edit other users, access raw data, add new gateways and sensors to the system, and check the system status.

4.2.2. Data management server

This server gathers, manages, and interacts with sensor and user data. It utilizes multiple Docker containers to host Node.js applications and a PostgreSQL database. By distributing tasks across containers, the system maintains limited functionality even if one or more containers fail. This server has the following components:

- **PostgreSQL:** The database is used to store the data essential for the application to function, including user-related data, patient information, samples received, medication parameters, and uplink/downlink frames.

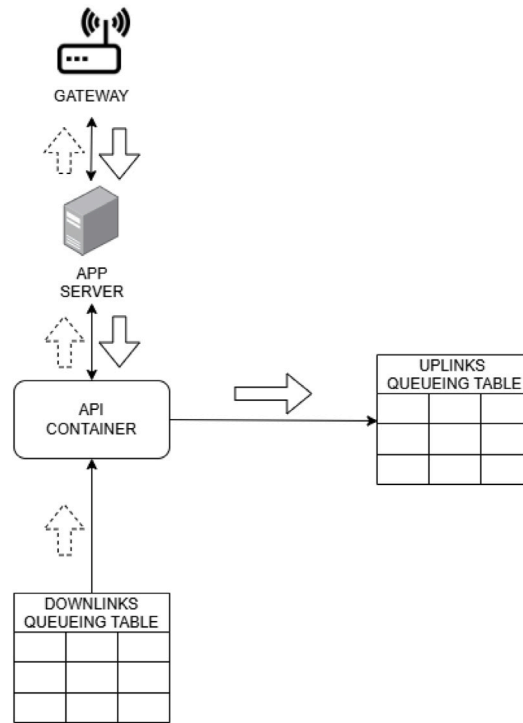


Fig. 6. Uplink/Downlink frames flow via WiFi.

- **Uplinks Container:** This container manages incoming uplink frames. Its principal function is processing these uplink frames, which involves decoding and classifying the data within the frames and subsequently storing them.
- **Downlinks Container:** Similar to the Uplinks Container, this container is responsible for managing outgoing downlink frames.
- **API Container:** This container serves as middleware for the *App server* to access the *Data Management server*. Its primary function is to present the data stored in the database to the web application.
- **System Container:** This container runs an algorithm that checks for changes in the LED status of the gateways every minute. When a change is detected, the algorithm generates a notification to inform Web App users.

Uplink/Downlink frames management

The uplink/downlink frames exchange can be split into two different flows: the WiFi gathering flow and the LoRa gathering flow. This process is supported by two tables in the PostgreSQL database, designed to temporarily store the uplink frames upon arrival and the downlink frames before leaving the system. These tables are called the *Uplinks Queueing Table* and *Downlinks Queueing Table*, respectively.

When a gateway is connected via WiFi, the communication of uplink/downlink frames follows the diagram in Fig. 6. Uplink frames ($\Rightarrow$) are sent to the *App server*. They are then forwarded to the API Container, which stores them in the Uplinks Queueing Table. Downlink frames ($\Leftarrow$) are periodically sent following the reverse path.

When a gateway uses LoRa, the uplink/downlink frames flow follows the diagram in Fig. 7. Uplink frames ($\Rightarrow$) are sent to the LoRa Server. These uplink frames are periodically retrieved by the Uplinks Container, which stores them in the Uplinks Queueing Table. Downlink frames ($\Leftarrow$) are periodically gathered by the Downlinks Container and subsequently transmitted to the LoRa Server, which then forwards them using LoRa.

To manage uplink/downlink frames regardless of the communication method used (WiFi or LoRa), the data follows the management procedure represented in Fig. 8. In this case, the arrows ($\Rightarrow$) represent the data flow. Every *queueing tables management period* (set to three minutes), all uplink frames stored in the Uplinks Queueing Table are gathered by the Uplinks Container. In the Uplinks Container, uplink frames are decoded, and if they contain data that needs to be stored (such as a sample), it is stored in the corresponding permanent table in the PostgreSQL database. If the uplink frame requires a response downlink frame (for example, if a new sensor needs to be activated), the Uplinks Container instructs the Downlinks Container to generate a new downlink frame, which will be stored in the Downlinks Queueing Table until it is sent to the corresponding gateway.

4.3. Data security and privacy considerations

Given the sensitive context of pharmaceutical use, the proposed system integrates multiple security mechanisms to protect data while enabling effective medication monitoring. When WiFi is available, communication between gateways and the central server

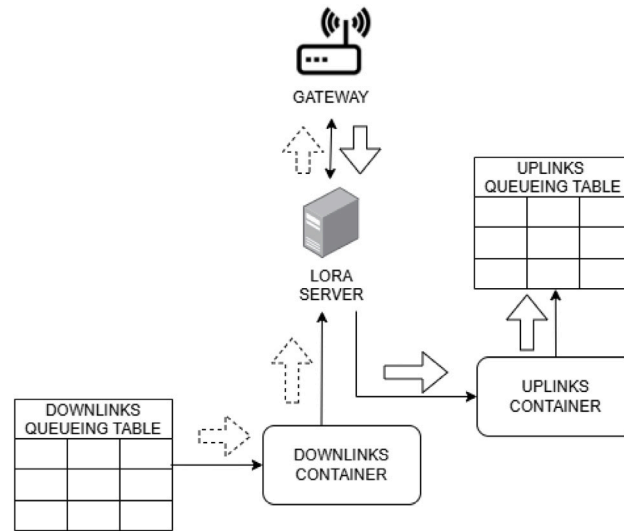


Fig. 7. Uplink/Downlink frames flow via LoRa.

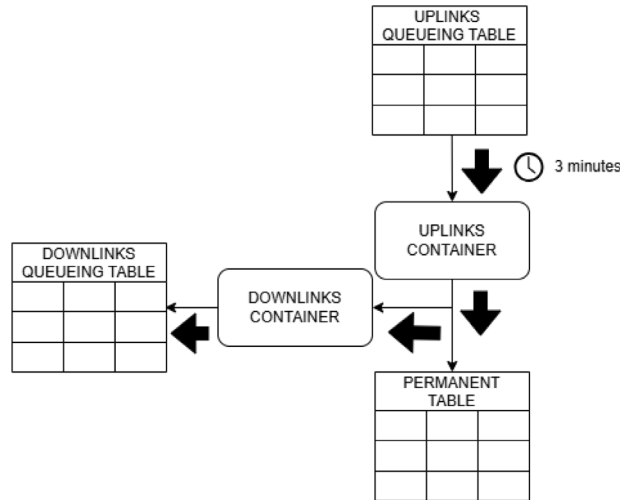


Fig. 8. Uplink/Downlink queueing tables management.

occurs over encrypted HTTPS channels. For LoRa-based communication, the system relies on LoRaWAN's native security features, including AES-128 encryption. Devices are provisioned with a unique AppKey and DevEUI for authentication. Additionally, the packet structure has been specifically designed for this system, making the data inherently difficult to interpret by unauthorized parties.

Importantly, the sensed parameters collected at patients' homes relate strictly to environmental conditions – such as temperature, humidity, UV exposure, and mechanical stress – and are used solely to assess the conservation quality of medications. The system does not collect or process any form of personal or behavioral data from patients.

To further protect patient identity, the linkage between a patient and a medication package is managed exclusively by the hospital pharmacy department. Each patient is assigned a unique code maintained locally by pharmacy staff, and this information is never stored or transmitted within the platform.

In addition, each sensor device is initially configured with a default password, which is changed upon deployment to a medication. Only the gateway and central server – via secure coordination – have access to the updated credentials required to reconfigure the sensor, such as switching its mode from datalogger to broadcast. This measure prevents unauthorized access and manipulation of sensor settings.

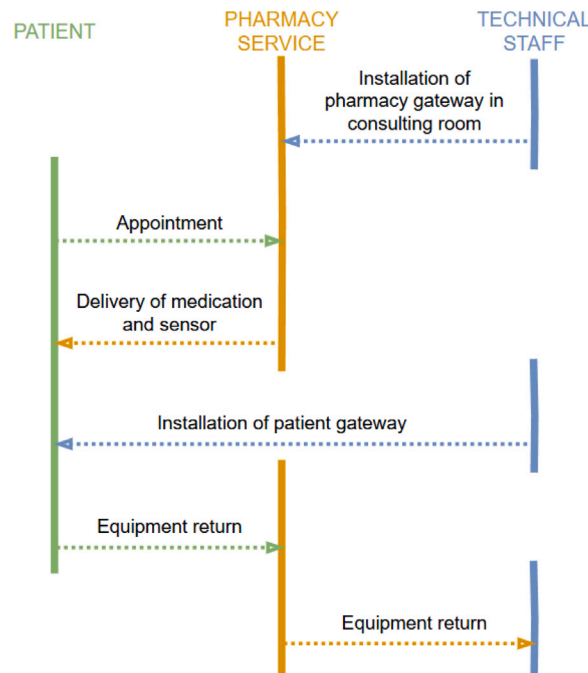


Fig. 9. Operation diagram.

5. System operation

This section provides an overview of the system's operational aspects, including a detailed description of the protocol and its implementation. It also covers the configuration process for the sensors and the operation of the *patient* gateway. Additionally, the role of the *App server* in setting up medications and patients, as well as monitoring the medicines, is illustrated.

5.1. Protocol overview and implementation

The protocol begins when a patient visits the hospital pharmacy to collect medication for several months. After consenting to participate, the pharmacist registers the patient via the project's website and assigns them a pre-registered sensor and *patient* gateway. The *pharmacy* gateway, installed in the consultation room (Fig. 9), then configures the sensor in *datalogger* mode. This mode ensures continuous environmental data recording from the pharmacy to the patient's home.

At home, the patient stores the medication with the attached sensor in the refrigerator. The technical team then installs the pre-registered *patient* gateway, either configuring it for WiFi (with patient consent) or using LoRa if coverage is available (Fig. 9). Once installed, the gateway connects to the server, which recognizes the pre-assigned sensor and returns configuration data. From that point, the gateway periodically scans the sensor—set to measure every *sensing time* (10 s)—and every *sending time* (10 min) sends summary statistics (average, max, min) to the *Data Management server* for monitoring and analysis.

To ensure adaptability, the system allows manual prioritization of WiFi or LoRa based on installation context. WiFi is preferred for its speed and reliability when available and permitted by the patient. Otherwise, LoRa is used by default. Regardless of the priority, both modes remain available, ensuring robust communication. This dual-mode setup is particularly advantageous in rural areas, where LoRa often provides better coverage due to the absence of signal obstacles such as dense buildings.

Each medication has defined temperature thresholds for safe storage: an ideal range (typically 2 °C to 8 °C) and absolute limits (usually 0 °C to 25 °C). Exceeding absolute limits is considered harmful and suspends administration until pharmacy review. Temperatures between the ideal and absolute thresholds are tracked, with each medication having a maximum allowable exposure time in this suboptimal range to ensure safety and efficacy.

The *patient* gateway not only transmits data to the server for pharmacy monitoring but also provides real-time alerts to patients in case of potential risks. It features visual and audio signals, coded to indicate whether action is needed or medication intake should be paused. As shown in Fig. 10, blue transitions occur automatically, while orange ones are triggered by pharmacy staff. Alerts are based solely on temperature readings to enable timely patient intervention against suboptimal storage conditions.

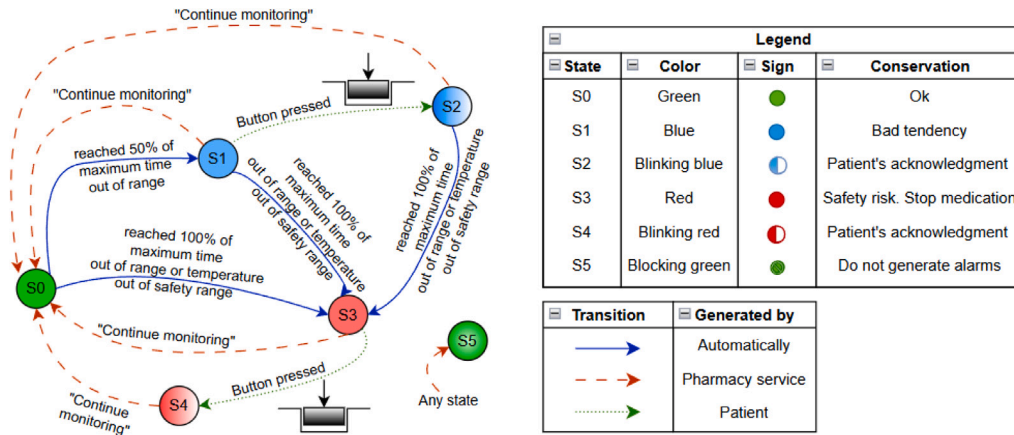


Fig. 10. Patient gateway states.

5.2. Sensor configuration

The sensor operates in two modes: *datalogger* and *broadcasting*. In *datalogger* mode, it records environmental data internally at regular intervals, starting when configured by the *pharmacy* gateway before the patient receives the medication. This allows continuous monitoring during transport from the hospital to the patient's home. Once at home, the patient places the medication and sensor in the refrigerator, where the sensor continues logging until the *patient* gateway is installed.

During installation, the gateway connects once to the sensor to switch it to *broadcasting* mode. In this mode, the sensor emits a Data Packet every *sensing time* (10 s), simultaneously taking new measurements. The gateway collects these values continuously over a *sending time* period (10 min), computes average, minimum, and maximum values from approximately 60 samples, and sends a summary message (which will be introduced in Section 5.4) to the *Data Management server*.

Ensuring data quality in IoT systems is essential for accurate monitoring and decision-making. This work incorporates data filtering and cleaning techniques to eliminate noise, outliers, or erroneous sensor readings before transmission. Two types of measurements are handled differently: periodic (temperature and humidity) and event-based (light radiation and mechanical stress).

- **Periodic data** is collected at a frequency determined by the *sensing period* (10 s) and summarized at a frequency determined by the *sending period* (10 min). If all measurements are correct, each 10 min window includes 60 samples, which are filtered using a moving average (to smooth out fluctuations and eliminate noise) and a median filter (to replace each value in a set by the median of the adjacent values). The final reported value is the average of the outputs from both filters, improving robustness.
- **Event-based data** is only generated during events. UV radiation is typically zero and only sampled during exposure, with the average intensity and sample count recorded to estimate duration. The accelerometer remains idle until motion is detected; when triggered, it records the maximum acceleration and number of events per *sending period*. Short bursts (<1 s) are classified as drops, while longer ones are labeled as shakes, enabling a distinction between transportation incidents and handling errors.

5.3. App server operation: Medicine and patient setup

The main functionality of the App is to monitor medication. All medicines are pre-added to the system, and pharmacists must follow a specific process when dispensing them to ensure data collection.

The process begins with patient registration. The pharmacy specialist assigns a unique patient code, links a gateway device, and defines the buzzer mute period using a dedicated form in the system interface. After registration, the specialist proceeds to associate the prescribed medication with the patient through a multi-step procedure described below.

i. Assign a Medicine to the Patient:

Using the system interface, the pharmacy specialist selects the room, identifies the patient, specifies the prescribed medication, and sets the administration time.

ii. Detect Sensor:

In the medicine delivery room, a *pharmacy* gateway detects and configures sensors (see Section 5.2). After step one, the pharmacy specialist powers on the sensor, allowing the gateway to detect it and send its data to the web App.

iii. Configure and Deliver the Sensor:

Once the sensor is detected, the system displays its location, last detection time, status, and provides an option to configure it. Activating the configuration sets the sensor to *datalogger* mode, initiating data sampling from that point onward.

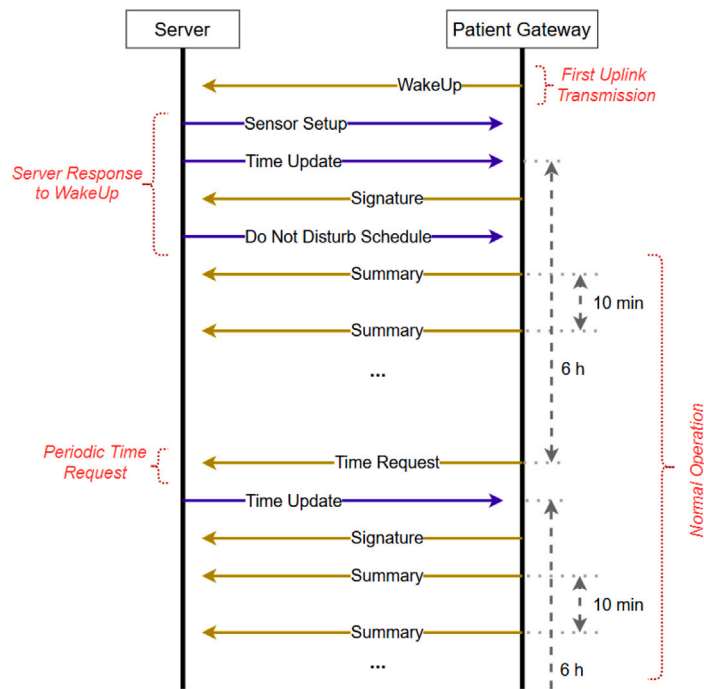


Fig. 11. Patient gateway operation.

5.4. Patient gateway operation

The *patient* gateway operation, illustrated in Fig. 11, is as follows:

- **Gateway Installation:** A technician installs the gateway at the patient's home, updating the algorithm with necessary variables – such as WiFi credentials, if applicable – before uploading it to the device. The gateway is then plugged in, ideally near the refrigerator in the kitchen.
- **Initialization and Network Communication:** Upon startup, the gateway attempts to initialize communication via the preferred method (LoRa or WiFi). If unsuccessful, it automatically switches to the alternative.
- **First Uplink Transmission:** Once communication is established, the gateway sends a one-time WakeUp signal to inform the server of its activation. Repeated WakeUps may indicate a power cycle due to unplugging or an outage.
- **Server Response to WakeUp:** When the server receives a WakeUp signal, it replies with three downlink messages: Sensor Setup, Time Update, and Do Not Disturb Schedule. These messages provide the gateway with the assigned sensor ID, current date and time, and the patient's preferred quiet hours, as configured during the pharmacy consultation. Details of each message are provided below.
- **Receiving Sensor Setup Message:** Upon receiving this message, the gateway retrieves the sensor's MAC address and creates a corresponding object containing its ID assigned by the server in the web application, temperature thresholds, password, and allowed exceedance time. It also stores the last recorded accumulated time outside optimal conditions to ensure continuity in case of resets caused by power interruptions.
- **Time Update Message:** This message provides the gateway with the current date and time, which it uses to synchronize its internal clock. Although the gateway tracks time locally thereafter, its ESP32-based hardware is susceptible to drift (discussed later). To confirm receipt, the gateway encrypts the Time Update Message with its private key and sends it back as a Signature Message. The *Data Management server* decrypts it using the public key. A match confirms successful authentication and ensures the response came from the authorized gateway.
- **Do Not Disturb Schedule Message:** This message communicates the patient's preferred quiet hours, recorded during their pharmacy visit. Sent by the server after the WakeUp, it ensures that any alarms occurring during this period are silenced by the gateway.
- **BLE Scanning and Connection:** After receiving the server responses, the gateway scans for the sensor's MAC address via BLE and connects using the password from the Sensor Setup Message. It retrieves all historical data recorded since pharmacy configuration and then switches the sensor to broadcast mode, allowing real-time BLE data transmission without maintaining a continuous connection.

	Type	ID sensor	Seq Num	% Bat	Mean Temp	Min Temp	Max Temp	Mean Humidity	Min Humidity	Max Humidity	UV Radiation	Acceleration	State	Current accumulated	Max accumulated
Summary	11	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Bytes	1	2	2	1	2	2	2	1	1	1	1	1	1	2	2

Fig. 12. summary message.

- **Normal Operation:** Once in broadcast mode, the gateway enters its standard operation, scanning for BLE broadcasts from the assigned sensor. Upon detecting a signal, it requests a Scan Response containing current measurements. During each *sending period* (10 min), the gateway gathers data and calculates the average, maximum, and minimum for temperature and humidity. These are compiled into a *summary message* (see Fig. 12) with *type* set to 1, indicating recent data. In contrast, *datalogger* mode messages use *type* 0, have lower priority, and are sent between *sending periods*.
- **Summary Message Transmission:** During normal operation, the gateway compiles a *summary message* (Fig. 12) every *sending period*, reporting average, maximum, and minimum temperature and humidity, UV radiation, acceleration data, and battery status. It also includes the time spent outside optimal conditions and the maximum allowable exposure time for the medication. Redundant elements help the server detect errors and trigger corrective actions. Each message includes the sensor ID, supporting multi-sensor monitoring per gateway.
- **Periodic Time Request:** To ensure accurate timekeeping and verify connectivity, the gateway sends a *Time Request Message* every *time update period*, set to six hours. The server replies with a *Time Update Message*, enabling time synchronization and confirming communication via LoRa or WiFi. If no response is received after three 30 min retries, the gateway assumes a connection loss and switches to *datalogger* mode, storing *summary messages* as *non-priority messages*. Upon reestablishing the connection, the gateway resumes normal operation and transmits both stored *non-priority messages* and new *priority messages*.
- **State Reporting:** Each *Time Request Message* includes a “status” field indicating the gateway’s current state, determined by the temperature readings from the sensor (see Fig. 10).
- **State Management:** While the gateway autonomously determines its operational state based on real-time sensor readings, it can also receive state updates from the server via a downlink *Alarm Message*. This dual mechanism ensures flexibility and robustness: the gateway can respond immediately to environmental changes, while the server retains supervisory control. Pharmacy staff, through the web application, are limited to triggering only a predefined subset of state transitions (see Fig. 10, orange transitions) to maintain safety and consistency. Crucially, even when disconnected from the server (e.g., due to network outages), the gateway continues monitoring the environmental conditions and updating its state locally based on both near real-time sensor input and values stored in *datalogger* mode. This local autonomy allows the gateway to issue alerts without delay, thereby preserving the near real-time nature of the system regardless of whether communication occurs over Wi-Fi or LoRa. This design ensures that poor storage conditions are detected and addressed promptly, independent of connectivity constraints.

5.5. App server operation: monitoring medicine

Once the gateway is installed at the patient’s home, authorized users (*Pharmacy specialists* and *Admins*) can begin monitoring the medication’s status. The monitoring interface presents key information, including the patient code, current alarm status, control buttons to send alerts to the gateway, medication name, scheduled administration date, current state, latest temperature reading, timestamp of the last measurement, and the percentage of time the medication has remained outside the prescribed conservation range (see Fig. 13).

Users can access a graph of historical data from this card, which lets them choose a start date and the number of days to display.

The pharmaceutical team receives environmental data from the deployed nodes and uses predefined thresholds, aligned with drug-specific conservation requirements, to assess potential risks. In cases of deviation (e.g., prolonged exposure to out-of-range temperatures), alerts can trigger patient outreach or prescription review, helping prevent ineffective or unsafe medication use.

6. Results and discussion

The proposed QChainMED platform is currently being deployed at the *Hospital Universitari Son Espases* in Palma, Spain. This platform aims to enhance the quality of medicine conservation in patients’ homes, recognizing that household refrigeration

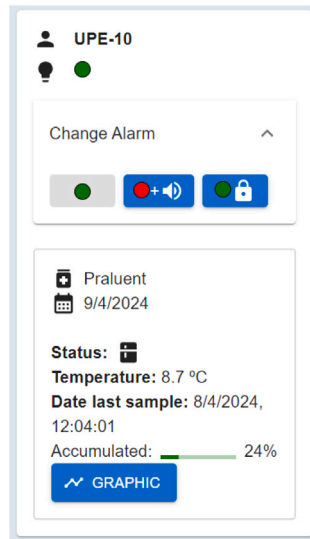


Fig. 13. Monitoring card.

equipment does not meet the same stringent requirements for temperature control as hospital systems. Additionally, due to the increasing impact of climate change in the Mediterranean region, medicines are more exposed to extreme temperature fluctuations. Therefore, the results presented below primarily focus on temperature as a critical factor in drug conservation. However, similar findings could be obtained for other monitored parameters – humidity, UV radiation, and mechanical stress – using the proposed sensor nodes.

6.1. Sensor node behavior discussion

Firstly, this section focuses on the results obtained from the designed sensor node, aiming to evaluate its capabilities for monitoring the storage conditions of medicines.

6.1.1. Sensor nodes battery lifetime estimation

As previously mentioned, minimizing energy consumption in the sensor nodes is a critical requirement. The sensor nodes are powered by a coin cell battery (CR2032) with a nominal capacity of 220 mAh, which should be sufficient to monitor the medical product until it has been administered.

The battery lifetime (*BLT*) under typical operating conditions can be estimated using Eq. (1). In this equation, the battery capacity is represented by its total charge storage, Q_{TOT} , and $\dot{Q}_{RATE}$ denotes the charge rate (i.e., the rate at which the battery is drained during operation). The lifetime is calculated as the ratio of the total charge storage to the charge rate:

$$BLT = \frac{Q_{TOT}}{\dot{Q}_{RATE}} \quad (1)$$

The charge drawn by a circuit from its power source (in this case, the battery) can be determined by integrating the consumption current. In our experimental setup, the consumption current was measured using the Power Profiler Kit II (PPK2) and the Power Profiler App, both provided by Nordic Semiconductor [29]. For example, Fig. 14 illustrates the consumption current curve of the sensor during a BLE broadcasting operation. The Power Profiler App directly calculates the charge drawn from the battery over a fixed time interval, which, in the case shown in Fig. 14, amounts to 21.6 μ C.

Table 3 presents the charge rate drained from the battery, based on charge measurements obtained using the PPK2. It is important to note that the base current – drawn when the sensor node is idle in DeepSleep mode (Base Current in Table 3) – makes a significant contribution to the total drained charge. Although other tasks may require higher current consumption, their brief duration relative to the long duty cycles means that power consumption during the idle state contributes significantly to the total energy usage.

Based on the experimental values presented in Table 3, and assuming a pessimistic estimate of $n = 10$ (since the medicine and the adjacent sensor node will rarely be subjected to movement within the refrigerator), the battery lifetime is estimated to exceed 35 months when applying Eq. (1). Even accounting for potential variations of up to 50%—due to factors such as differences in power consumption between sensor nodes, increased transmission power from environmental conditions in the patient's home, battery leakage, and deviations in battery capacity from the nominal value—the resulting battery runtime of 17.5 months would still be more than adequate for our application.

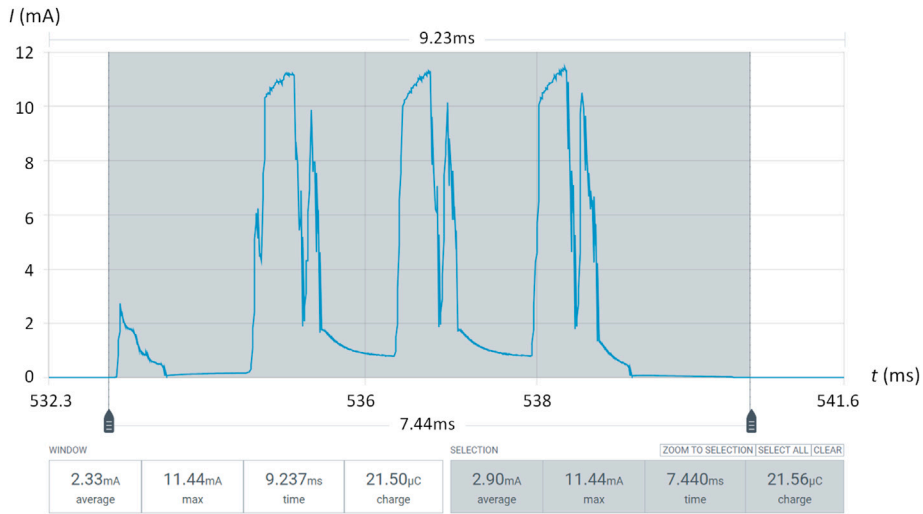


Fig. 14. Current drawn from the battery during a BLE broadcasting, measured using Power Profiler Kit II.

Table 3

Charge rate drained by the sensor node from the battery ($\dot{Q}_{\text{RATE}}$ in $\mu\text{C/s}$). One temperature and humidity measurement, and one broadcasting each 10 s. Accelerometer and light sensor are assumed to be triggered n times per day.

BLE broadcast	2.16
Temperature & humidity acquisition	1.84
Base current (Deep sleep)	4.24
Accelerometer and light sensors	$2380 \frac{n}{3600 \cdot 24}$
Total $\dot{Q}_{\text{RATE}}$	$8.24 + \frac{n}{36.30}$

6.1.2. Sensor nodes communications coverage estimation

To assess the impact of gateway positioning, sensor node placement within the refrigerator, and frequency band occupancy on BLE connectivity, we performed a series of measurements under controlled conditions.

The refrigerator is located in a non-rectangular room with an area of approximately 40 m². The study considered two different gateway positions relative to the refrigerator:

- **Location 1:** Located at a distance of 6 m, a 45° angle to an imaginary perpendicular extending from the refrigerator door.
- **Location 2:** Located at a distance of 7 m, a 135° angle to an imaginary perpendicular extending from the refrigerator door.

Additionally, the sensor node was placed in three different positions within the refrigerator:

- **Position (a):** On the central shelf of the door.
- **Position (b):** On the upper shelf, surrounded by filled bottles.
- **Position (c):** On the central shelf, with minimal surrounding food items.

To assess the influence of Bluetooth frequency occupancy, measurements were taken during two time intervals:

- **High-occupancy time slot (14:00),** when numerous nearby Bluetooth devices were active.
- **Low-occupancy time slot (03:00),** with significantly fewer active Bluetooth devices.

Each measurement session lasted 10 min. Since the sensor node transmitted a BLE packet every 10 s, the maximum number of successful transmissions per session was 60. The connectivity performance was expressed as the percentage of successful connections relative to this maximum value. Measurements were performed in five consecutive 10 min intervals for a minimum of three days. For each configuration, we report the minimum, maximum, and average percentage of successful connections.

Table 4 presents the results for the high-occupancy time slot (14:00), and Table 5 presents the results for the low-occupancy time slot (03:00). Using this data, we can analyze how sensor placement and Bluetooth occupancy (time slot) influenced success rates at each gateway location. The analysis shows that Bluetooth congestion (occupancy) was the primary factor limiting BLE connectivity. However, gateway placement played a crucial role in determining the extent to which sensor position affected performance. To optimize BLE performance, the following actions can be taken:

Table 4
Successful connection rates (%) during high-occupancy time slot (14:00).

Configuration		Success rate (%)		
Sensor position ^a	Gateway location ^a	Avg	Min	Max
Door (a)	Location 1	85.6	78.3	92.7
	Location 2	86.5	65.0	98.3
Upper shelf (b)	Location 1	86.4	51.7	98.3
	Location 2	76.2	58.3	91.7
Central shelf (c)	Location 1	86.4	58.3	98.3
	Location 2	67.0	55.0	85.0

^a See text for description of sensor positions and gateway locations.

Table 5
Successful connection rates (%) during low-occupancy time slot (03:00).

Configuration		Success rate (%)		
Sensor position ^a	Gateway location ^a	Avg	Min	Max
Door (a)	Location 1	96.1	88.3	98.3
	Location 2	92.2	78.3	98.3
Upper shelf (b)	Location 1	95.5	91.7	98.3
	Location 2	90.4	71.7	98.3
Central shelf (c)	Location 1	95.7	88.3	98.3
	Location 2	93.7	85.0	98.3

^a See text for description of sensor positions and gateway locations.

- Minimize interference from other Bluetooth devices. However, it is important to note that occupancy levels in a domestic environment are typically much lower than those in the room where the experiment was conducted (a university laboratory).
- Strategically place the gateway to minimize the impact of physical obstacles.
- If the gateway cannot be positioned optimally, sensor placement becomes more critical. Avoid placing sensors near materials that block signals (e.g., liquid-filled bottles).

6.1.3. Maximum capacity estimation in datalogger mode

The *Sensor Node* is initialized in datalogger mode when there is no connection to the *Gateway*. In this mode, data is stored at regular intervals defined by the *sending period*. Temperature and relative humidity values are stored as 2-byte variables, with a resolution of 0.1 °C for temperature and 1% RH for humidity. The UV radiation data is stored using 2 bytes: the first byte represents the number of times UV radiation was detected during the *sending period*, and the second byte stores the maximum UV radiation value recorded. Mechanical stress is quantified by the perceived acceleration, stored as 2 bytes: the first byte counts the number of times mechanical acceleration was detected, and the second byte indicates the maximum measured acceleration. Therefore, in *Datalogger mode*, a total of 8 bytes are used to store the variables at intervals defined by the *sending period*.

The *Sensor Node* is implemented using the nRF52840 microcontroller in *Datalogger mode*, where the data is stored in memory. The maximum memory capacity in *Datalogger mode* is set to 128 KB, allowing for the storage of approximately 16 384 records, based on the described format. Given that the *sending period* is assumed to be 10 min, the sensor node can store measurements for a total of 113 days and 18 h—providing a substantial margin for the application.

6.2. Data analysis

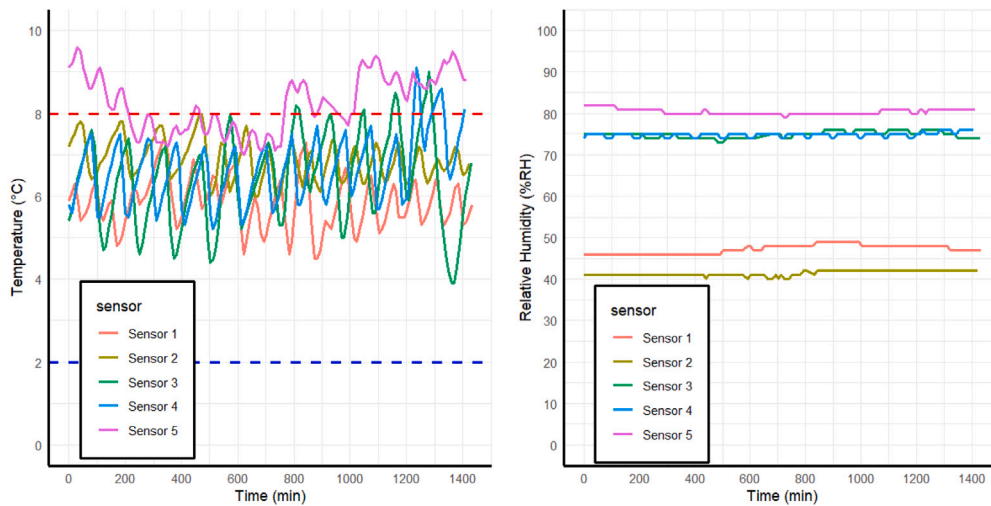
In this section, the overall behavior of the QChainMED platform will be analyzed, along with the results obtained from laboratory experimentation.

6.2.1. Effect of refrigerator behavior

During the experimental phase, and in line with previous studies [7], the temperatures recorded inside the refrigerator were not constant but fluctuated. Fig. 15(a) illustrates the temperature variations registered by five different sensors used by various patients. As shown in the plot, the temperature typically follows a cyclical pattern. This cyclical behavior is directly related to the refrigerator's hysteresis cycle, in which the refrigerator compressor turns on and off based on the thermostat setting. Several factors can influence the cycle, including the time of year, the frequency with which the refrigerator door is opened or closed, and the placement of the medication inside the fridge. The plot in Fig. 15(a) shows data from several sensors, with the majority of measurements remaining within the temperature range deemed suitable for optimal medicine conservation.

Thus, depending on the refrigerator model and design, the temperature cycles may vary in intensity.

In contrast, the humidity variable was not used in this study to assess whether a medicine is being properly preserved. However, it is stored in the platform in the same manner as temperature. For the same sensors shown in Fig. 15(a), the relative humidity



(a) Temperature measured by different sensors. (b) Relative Humidity measured on a given fridge.

Fig. 15. Temporal behavior of temperature and relative humidity.

Table 6

Times obtained on the final implementation.

Task	Time in minutes
Publishing <i>Sensor Node</i> data	0.17
Sending <i>Gateway</i> data	10
Data reception from LoRaWAN server	16
Data reception using WiFi	1
Server message processing time	<114 ms
Server message to Gateway sending using LoRaWAN	<1214 ms

behavior is represented in Fig. 15(b). Unlike temperature, humidity remains more stable over time, and its values do not follow a cyclical pattern. Of course, relative humidity levels depend on factors such as the sensor's location inside the fridge, the type of refrigerator, and other uncontrollable variables.

Although this variable was not used to trigger patient warnings, it did raise concerns when relative humidity values exceeded 60% RH, as shown in Table 2. In Fig. 15(b), sensors 3, 4, and 5 exhibit values above 70% RH, clearly indicating undesirable exposure to high humidity. This could be attributed to the use of a refrigerator without a frost-free system, although this could not be confirmed through further analysis.

6.2.2. WiFi and LoRa processing times

Another important performance parameter of the system is the processing times of the implementation. These times have been analyzed to ensure that the system provides near real-time responses. Table 6 summarizes the key processing times for the tasks executed on the platform.

The values from the different sensors installed in patients' homes are processed on the platform at 16 min intervals. This time is set because the sensors transmit values to the LoRa network server every 10 min, allowing all received data to be collected while reducing the number of accesses to the IoTIB network server.

In contrast, if the gateway device communicates via the patient's WiFi network, data reception and processing occur at 1 min intervals. This shorter interval is possible because the reception by the gateway bypasses the need for processing through a network server (see Table 6 for more details).

This difference in processing frequency is also reflected in the message transmission path. When using an IP-based connection, messages are transmitted directly to the gateway, whereas with a LoRaWAN server-based connection, messages pass through an additional network layer. As a result, IP-based transmission has been observed to deliver messages up to 10 times more efficiently than LoRa.

6.2.3. Communications reliability

The two data transmission methods from the patient gateway have been compared. A gateway is expected to transmit 148 messages per day (144 summary messages and 4 time request messages). Fig. 16 presents a comparison of the number of messages received by the server over a one-week period (7 days) when the patient's gateway is connected via LoRa versus when it is connected via the patient's WiFi network.

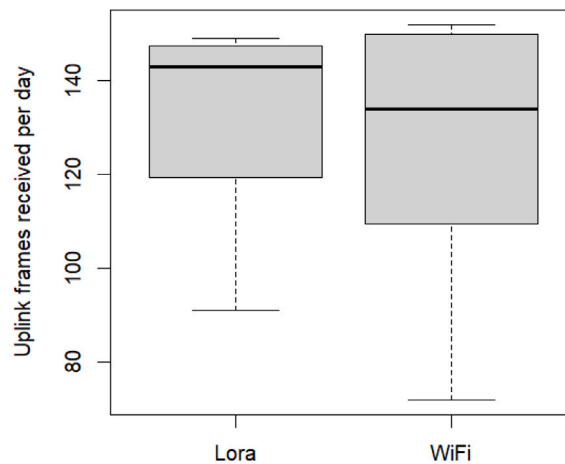


Fig. 16. Messages received via WiFi and LoRa during a week.

Observing the results, the WiFi connection achieves more days in which all packets successfully reach the platform. However, the average number of messages received per day is lower compared to the LoRa connection. Additionally, the WiFi connection exhibits days with significantly fewer received packets than the LoRa connection. These discrepancies may be attributed to the LoRa network server, which manages message loss and enables automatic retransmission. In contrast, with the WiFi connection, lost messages – whether due to Internet failures or delivery issues – lack a recovery mechanism, leading to greater variability in received data.

Overall, if the gateway has good LoRa coverage, data reception is generally more reliable compared to WiFi connections. However, the monitoring system is designed to be robust, ensuring that even if some data fails to reach the platform, the gateway still considers it when assessing the conservation status of the medicines. While occasional message loss may occur, the alert mechanism remains reliable, effectively detecting and responding to periods of inadequate storage conditions.

7. Conclusions

This research introduces an opportunity to enhance hospital care processes by enabling the remote monitoring of drug storage conditions within patients' homes. The ability to automatically track these conditions improves both healthcare resource management and patient safety. Additionally, the system actively engages patients in ensuring proper medication storage. Interaction is facilitated through intuitive visual and auditory alerts, as well as a dedicated button on the patient gateway—empowering patients to stay informed and respond promptly to any deviations.

We have presented the implementation of QChainMED, an IoT-based multi-sensor platform designed to monitor ambient-sensitive medications. The system architecture comprises two main components: a *monitoring network* and a *server system*. The monitoring network includes a custom sensor node and a patient gateway installed near the medication storage location. It enables automatic and remote monitoring of temperature, humidity, light radiation, and mechanical stress, while also performing local assessment of storage conditions. This edge-level processing supports autonomous alerting and patient engagement without requiring hospital-side intervention. Each prescribed medication course can thus be individually monitored, with automated alerts sent to both patients and pharmacy personnel. The server system complements this by managing data storage, alert handling, and interaction with pharmacy professionals.

The integration of multiple sensing variables enables a more comprehensive assessment of medicine conservation status, opening pathways for future correlation with drug efficacy. The use of BLE technology allows the sensor to interface with a broader range of devices, facilitating its adaptation to different deployment scenarios. Importantly, the gateway's ability to issue early warnings to patients represents a shift from traditional retrospective monitoring toward more proactive and preventive interventions.

Although the sensor node is still under development, the current implementation is already capable of transmitting data related to temperature, humidity, light radiation, and mechanical stress. While this manuscript focuses primarily on temperature and humidity – due to the availability of well-established conservation thresholds – the system also logs light and mechanical events and sends them to the central server. A detailed technical description of how these variables are handled can be found in our related work [26].

While the present work centers on the system's design, deployment, and technical validation, its broader impact lies in supporting clinical decision-making. The platform allows healthcare professionals to remotely assess storage conditions and identify potential risks before medication efficacy is compromised. As part of an ongoing clinical study, we are collecting feedback from both pharmacists and patients to evaluate how the platform informs real-world decisions, improves adherence, and reduces wastage. These findings will be reported in a forthcoming, clinically focused publication.

Currently, the gateway performs basic local processing, such as computing summary statistics and issuing alerts when conservation thresholds are breached. Future versions may incorporate more advanced, decentralized intelligence. For example, federated

learning (FL) techniques could enable gateways to collaboratively train models for anomaly detection or predictive maintenance – such as forecasting refrigeration failures – without transmitting sensitive raw data. This approach would enhance privacy, reduce bandwidth demands, and enable adaptive, privacy-preserving system behavior. As noted by M. A. Almaiah et al. [30], FL offers a promising framework for secure AI/ML deployment in healthcare environments.

It is also important to note that the system is designed for reuse. Both the sensor nodes and patient gateways are redeployed across different treatment cycles and patients. Sensors are typically placed inside the packaging containing the complete set of prescribed doses, allowing for continuous tracking over the entire treatment period. Future iterations will explore standardized, reusable enclosures – such as sleeve-style casings – to improve usability, durability, and scalability.

In conclusion, the QChainMED platform responds to the growing need for holistic solutions in healthcare by addressing not only the patient but also the treatment and its conservation conditions. This is particularly relevant in the era of personalized medicine, where biologic drugs are increasingly prevalent and highly sensitive to improper storage.

CRediT authorship contribution statement

Maria Frontera-Bergas: Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Miguel Vinaixa-Fernández:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Bartomeu Oliver-Riera:** Writing – original draft, Investigation, Formal analysis. **Jaume Ramis-Bibiloni:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Eugeni Isern:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Bartomeu Alorda-Ladaria:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used OpenAI ChatGPT exclusively in order to improve readability and language of the work. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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